

New York City Accident Data Analysis

Using SQL & Python

Project Overview

This project focuses on analyzing motor vehicle collision data from New York City (January 2018 to August 2019) to identify patterns , trends , and key risk factors contributing to road accidents. The goal was to generate actionable insights that support data- driven decision-making aligned with the city's road safety initiative , *Vision Zero* aimed at reducing traffic-related fatalities.

Objective

The primary objective of this project was to :

- Analyze accident trends over time .
- Identify high-risk time periods and locations .
- Determine major causes of accidents .
- Understand the impact of vehicle types and borough-level differences .
- Provide insights to support preventive safety measures .

Tools & Technologies Used

- **SQL (Amazon Athena)** – Data querying and aggregation .
- **Python** – Visual Studio Code .
- **Pandas & NumPy** – Data cleaning, transformation , and analysis .
- **Data Visualization Libraries** – Matplotlib/Seaborn .

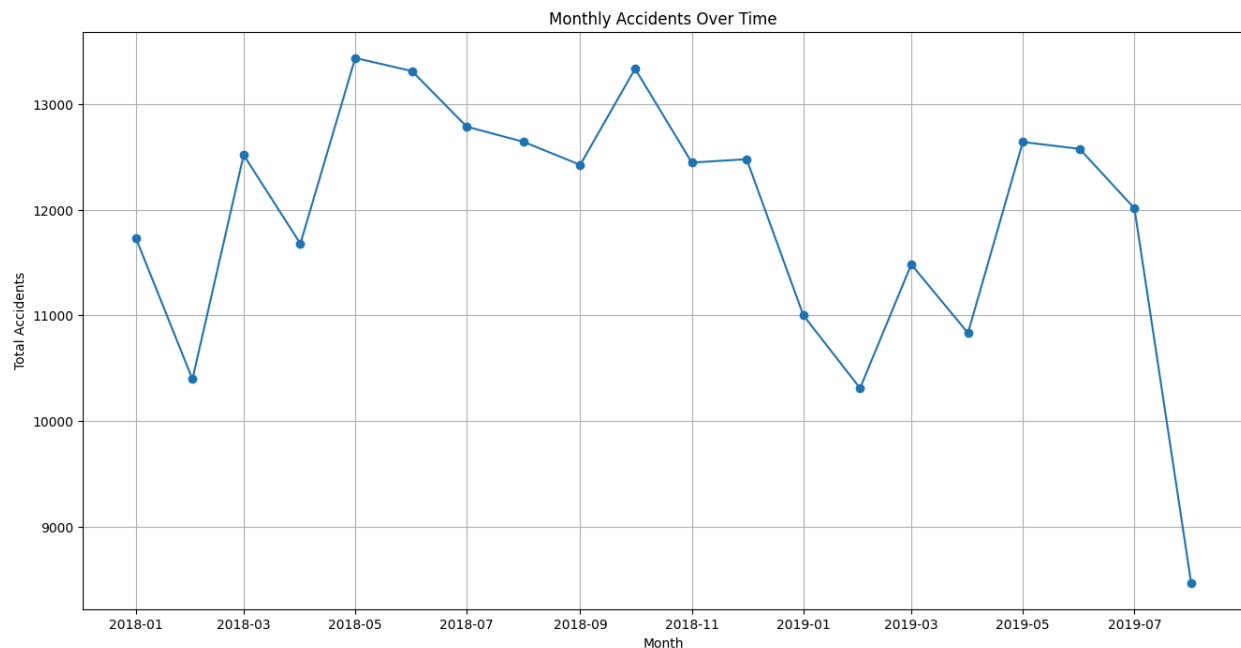
Data Preparation

- Imported and processed raw CSV data with a non-standard delimiter (;) using Python
- Cleaned missing and inconsistent values across key columns (borough , contributing factors, vehicle types)
- Extracted and merged additional borough-related data from a JSON file .
- Converted date and time fields into structured formats for time-series analysis .

Analysis & Key Insights

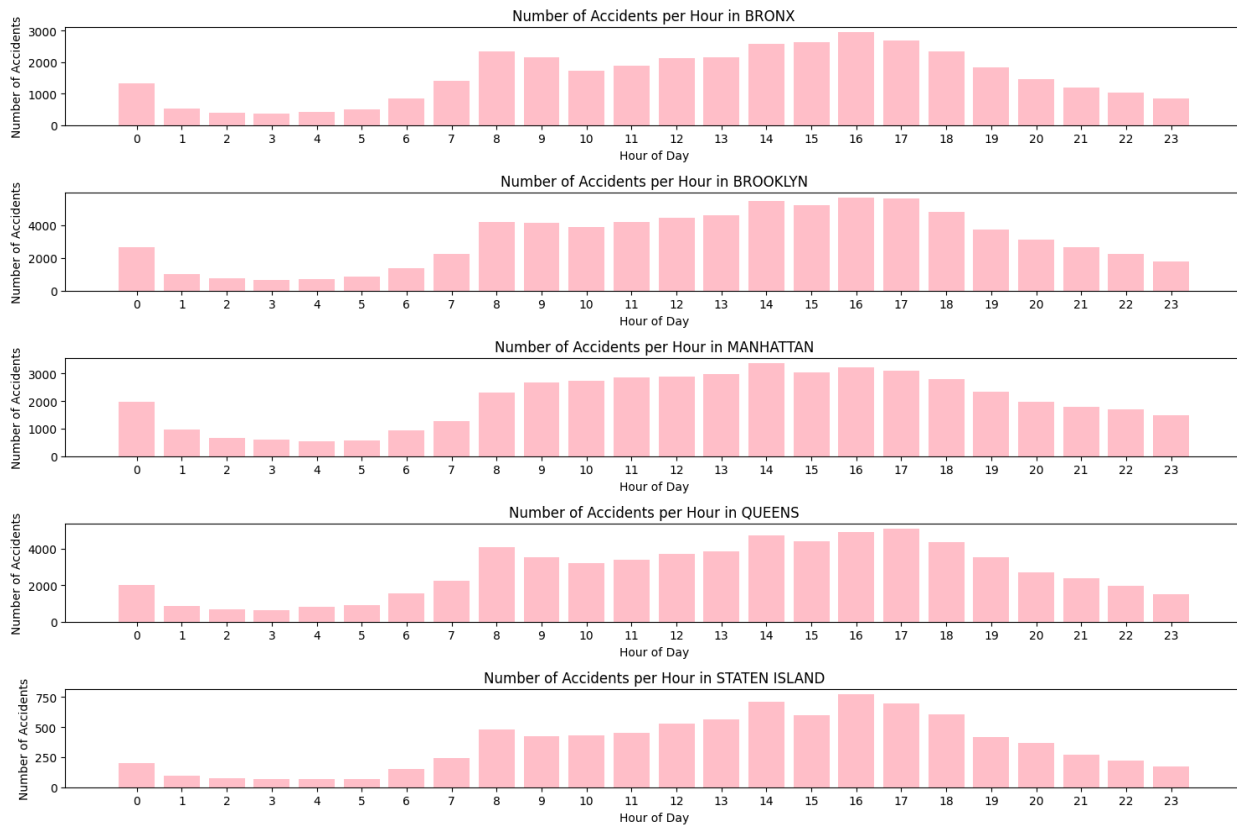
1. Trend Analysis

- Conducted time-series analysis to evaluate accident trends over 1.5 years
- Identified fluctuations and periods with increased accident frequency .



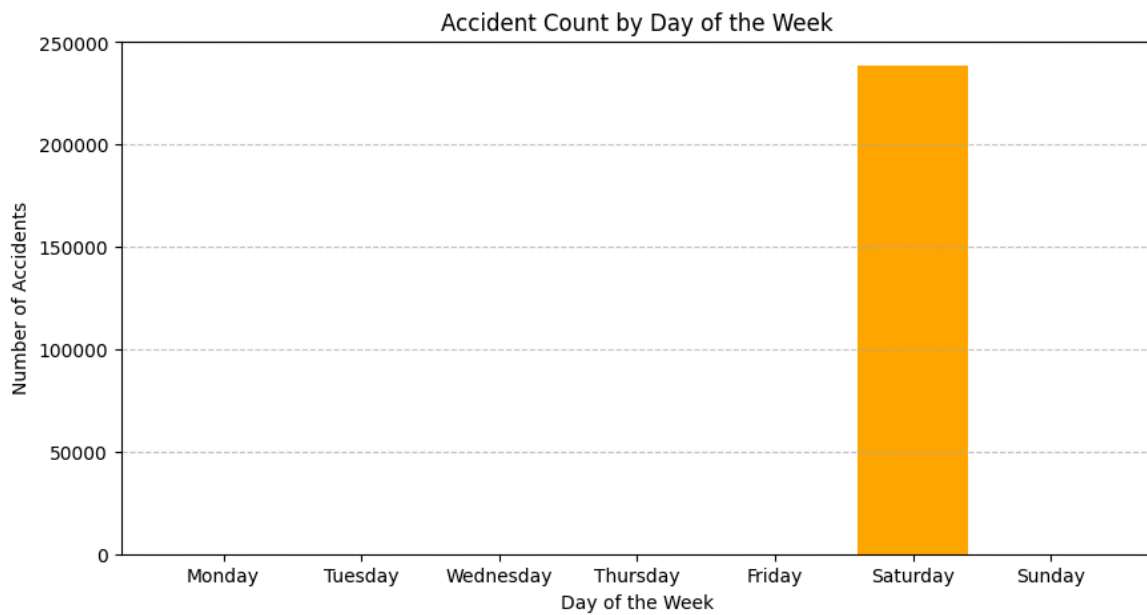
2. Time-Based Patterns

- Analyzed accidents by hour of the day .
- Found peak accident occurrences during **rush hours (morning and evening commute)**



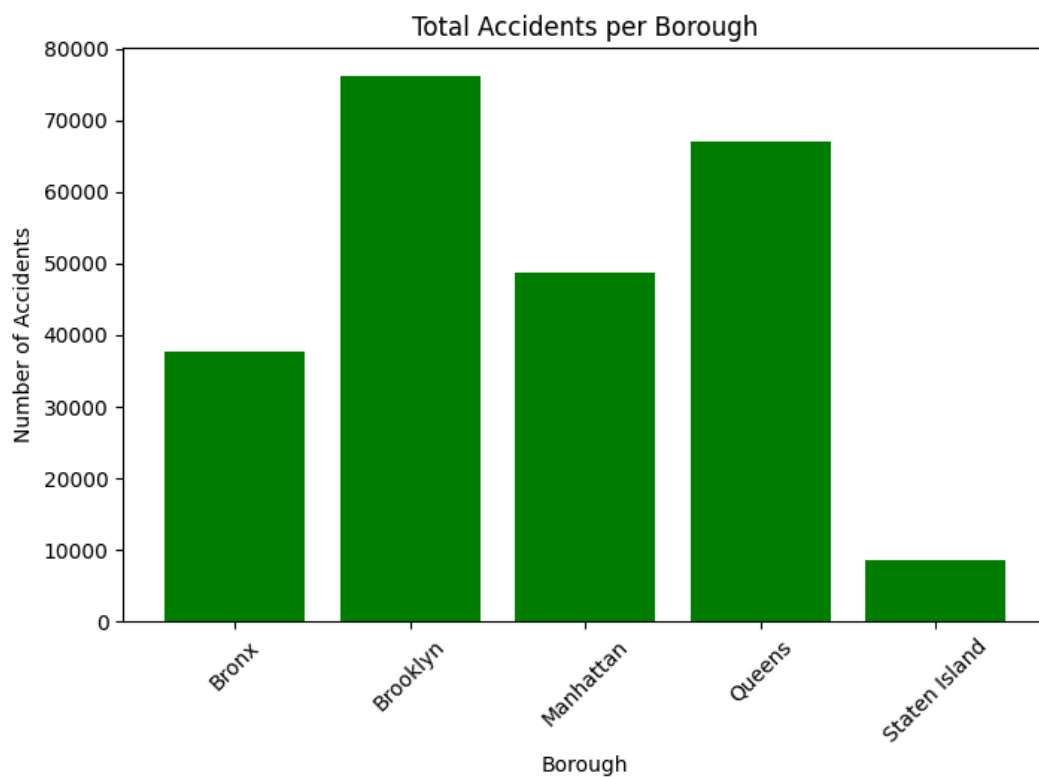
3. Weekday vs Weekend Analysis

- Compared accident frequency between weekdays and weekends
- Observed higher accident rates on weekend (Saturday) due to increased traffic volume



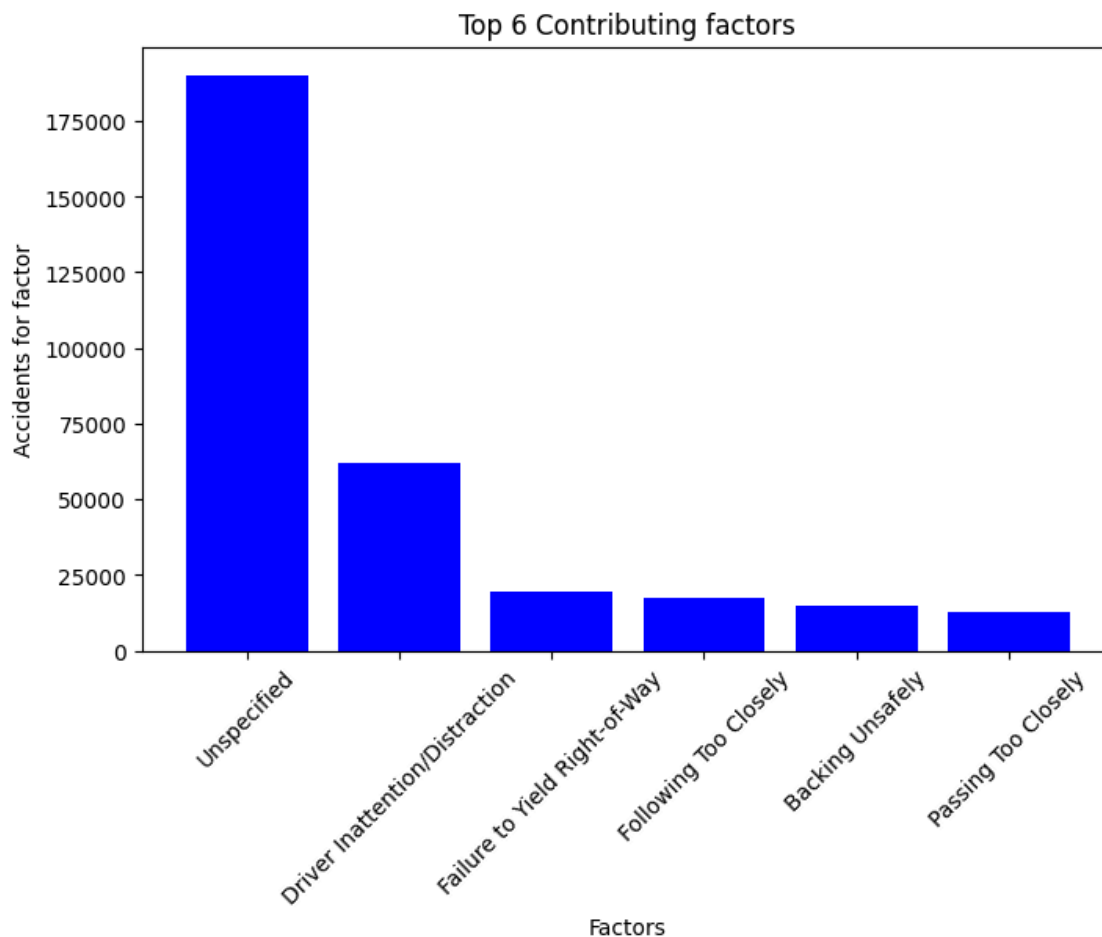
4. Borough-Level Analysis

- Calculated **accident density** (accidents per area)
- Identified boroughs with disproportionately high accident rates , highlighting high-risk zones



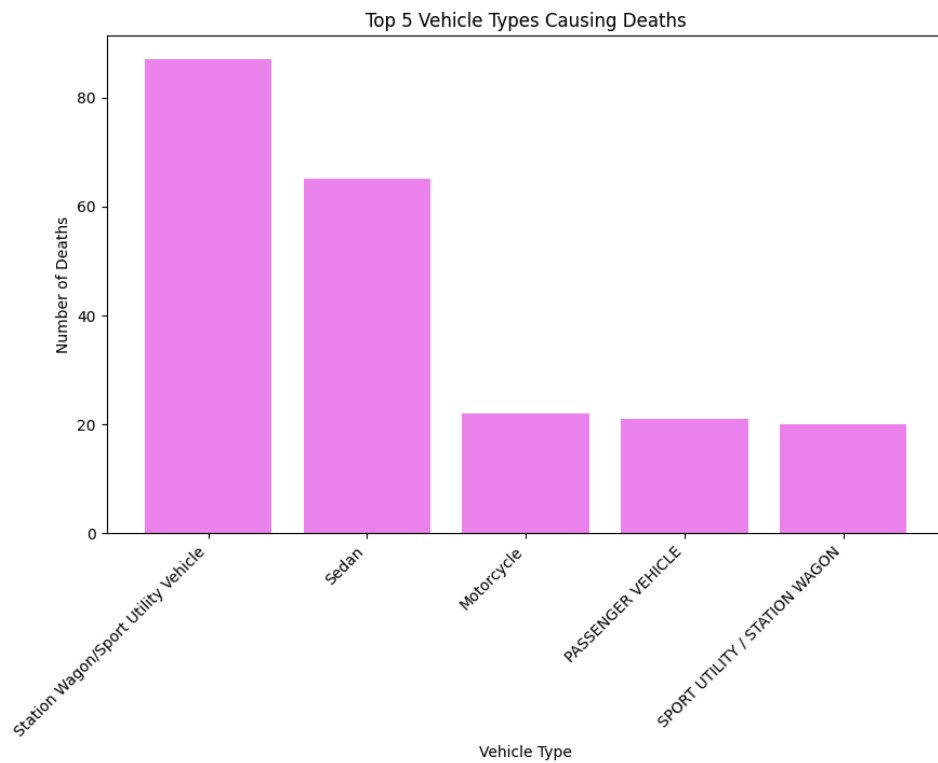
5. Cause of Accidents

- Extracted and ranked the **top 6 contributing factors** .
- Found driver-related behaviors (e.g., Distraction , speeding) as major causes



6. Vehicle Type Analysis

- Identified most frequently involved vehicle types across boroughs .
- Analyzed vehicle types associated with fatalities to determine high-risk categories



Business Impact :

- Provided insights to support **targeted road safety interventions**
- Helped identify **high-risk locations and time periods** for focused enforcement
- Supported decision-making for initiatives under *Vision Zero* , including :
 - Speed reduction programs
 - Road safety audits
 - Pedestrian and cyclist safety improvements
 - Public awareness campaigns

Conclusion :

This project demonstrates the use of **SQL and Python-based data analysis** to transform raw accident data into meaningful insights . The findings can assist policymakers and city planners in implementing targeted strategies to reduce accidents and improve urban road safety .